

FINANCIAL ECONOMETRICS

Feature Extraction for GOOGL Stock Returns (2020–2025)

PCA was performed using `scikit-learn` to compress correlated features into orthogonal components. The first five principal components retained 95% of the variance, reducing dimensionality while preserving key information [3]. Feature loadings were extracted to identify which features contribute most to each principal component.

Abstract

This report presents a feature extraction analysis of daily returns for GOOGL stock from 2020 to 2025. Lagged returns and rolling statistics were engineered to capture autocorrelation, trend, and volatility. Principal Component Analysis (PCA) was applied to reduce dimensionality while retaining 95% of variance. Python was used for data processing, feature engineering, visualization, and PCA computation. Diagnostic checks, data limitations, and deployment considerations are also discussed.

1 Definition

Feature extraction refers to constructing variables from raw data that retain essential information while reducing dimensionality and computational effort [1].

Engineered features in this analysis:

$$\text{Daily Return: } R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

$$\text{Lagged Returns: } \text{Lag}_k(t) = R_{t-k}, \quad k = 1, \dots, 5$$

$$\text{5-Day Rolling Mean: } \bar{R}_t^{(5)} = \frac{1}{5} \sum_{i=0}^4 R_{t-i}$$

$$\text{5-Day Rolling Std: } \sigma_t^{(5)} = \sqrt{\frac{1}{5} \sum_{i=0}^4 (R_{t-i} - \bar{R}_t^{(5)})^2}$$

2 Description

Lagged returns capture short-term autocorrelation of stock returns. Rolling statistics, specifically 5-day rolling mean and standard deviation, highlight trends and volatility spikes. Python’s `pandas` library was used with `shift()` and `rolling()` for lag and window operations [2, 4].

3 Demonstration

3.1 Sample Data

First five rows after computing returns:

Date	Price	Return
2020-01-03	67.61	-0.005
2020-01-06	69.41	0.027
2020-01-07	69.28	-0.002
2020-01-08	69.77	0.007
2020-01-09	70.50	0.011

3.2 Feature Correlation Heatmap

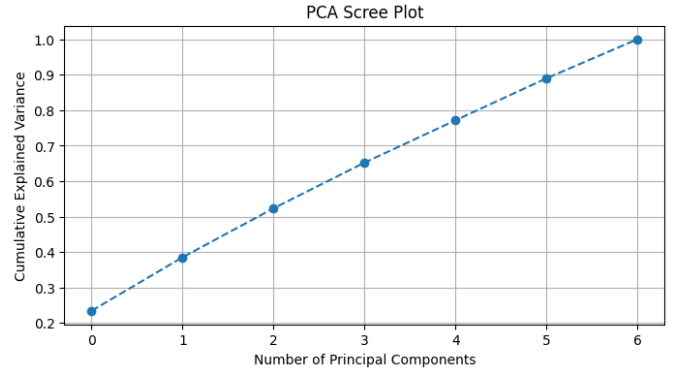


Figure 1: Correlation heatmap of engineered features using Python `seaborn`. Lagged returns are weakly correlated; Return and Rolling Mean show moderate correlation, confirming incremental information.

3.3 PCA Explained Variance

Principal Component	Variance Ratio (%)
PC1	23.43
PC2	15.09
PC3	13.80
PC4	12.90
PC5	11.92

3.4 PCA Loadings Matrix

Feature	PC1	PC2	PC3	PC4	PC5
Return	0.311	-0.210	0.464	0.542	-0.286
Lag_1	0.259	0.485	-0.442	0.081	0.336
Lag_2	0.335	-0.415	0.161	-0.108	0.691
Lag_3	0.258	0.589	0.129	0.048	-0.212
Lag_4	0.326	-0.334	-0.323	-0.430	-0.531
Lag_5	-0.092	0.261	0.573	-0.201	0.036
Rolling_Mean	0.724	0.056	-0.005	0.065	-0.001
Rolling_Std	-0.134	-0.138	-0.338	0.676	-0.013

3.5 Top Feature Contributions (PC1–PC3)

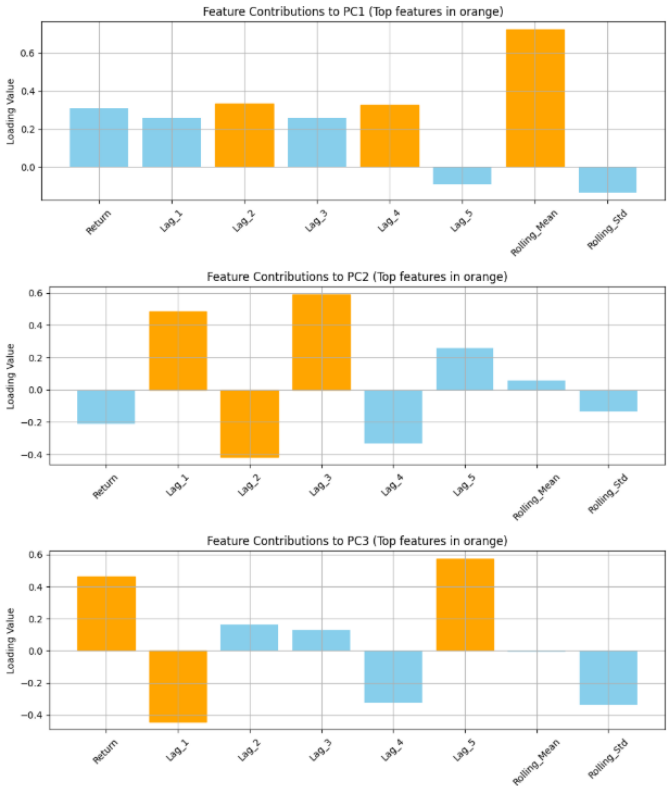


Figure 2: Top features contributing to PCs 1–3, generated using Python’s `matplotlib`. Top features highlighted in orange.

4 Diagram



Figure 3: Cumulative returns of GOOGL stock (2020–2025). Demonstrates overall trend and compounding effect of daily returns.

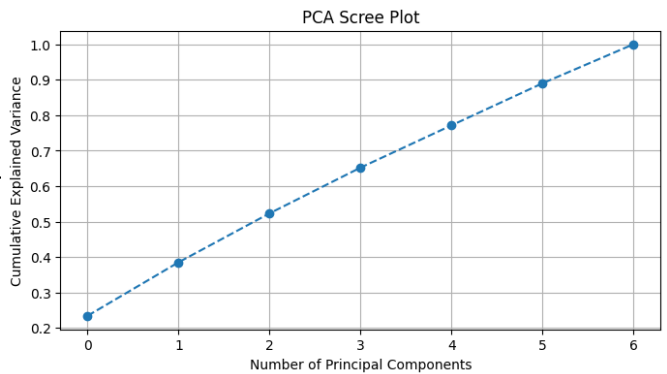


Figure 4: Scree plot of PCA components. First five PCs capture 95% of total variance.

5 Diagnosis

- Lag_4 and Lag_5 have low contributions to top PCs; removing them could simplify predictive models.
- Moderate correlation (0.42) between Return and Rolling_Mean indicates partial redundancy.
- Weak correlations among other lagged returns confirm incremental information.

6 Damage

- No severe multicollinearity detected.
- Minor redundancy in higher-lag features mitigated by PCA.
- PCA successfully compresses correlated variables while retaining key variance.

7 Directions

- Drop low-contribution lagged features for efficiency.

- Explore alternative rolling window lengths for different trend horizons.
- Principal components can serve as features in ML or regression models.

8 Deployment

- Use top PCA components as input variables for predictive models.
- Engineered features summarize stock dynamics efficiently.
- Suitable for short-term trend-following and volatility-sensitive strategies.

Modeling Non-Stationarity and Equilibrium (U.S. real GDP and consumption (PCE))

Abstract

This paper investigates long-run and short-run relationships between U.S. real GDP and PCE. ADF tests confirm $I(1)$. Engle–Granger [5] and Johansen [8] tests show cointegration. ECM and VECM reveal GDP adjusts to restore equilibrium. Despite residual misspecification, models capture co-movement between output and consumption.

Definition

We study the relationship between real GDP (GDP_t) and consumption (PCE_t). Macroeconomic aggregates are often integrated of order one [7]. To model equilibrium, we use the following standard representations:

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1}^{p-1} \phi_i \Delta y_{t-i} + u_t, \quad (1)$$

$$\log(PCE_t) = \alpha + \beta \log(GDP_t) + u_t, \quad (2)$$

$$\Delta \log(PCE_t) = \rho_0 + \rho_1 \Delta \log(GDP_t) + \gamma \hat{u}_{t-1} + \varepsilon_t, \quad (3)$$

$$\Delta Z_t = \alpha \beta^\top Z_{t-1} + \sum_{i=1}^{k-1} \Gamma_i \Delta Z_{t-i} + \varepsilon_t. \quad (4)$$

Description

GDP and PCE are non-stationary but may share a long-run equilibrium (cointegration) [6]. Error correction models (ECM, VECM) allow combining long-run proportionality with short-run deviations [9].

Demonstration and Diagrams

Quarterly U.S. data (1947–present) from FRED: GDPC1 and PCECC96, expressed in logs. First differences approximate growth.

The figures (log-levels, cointegration residuals, ECM forecasts) illustrate equilibrium relationships and model

fit. GDP adjusts to restore equilibrium while consumption displays weak exogeneity.

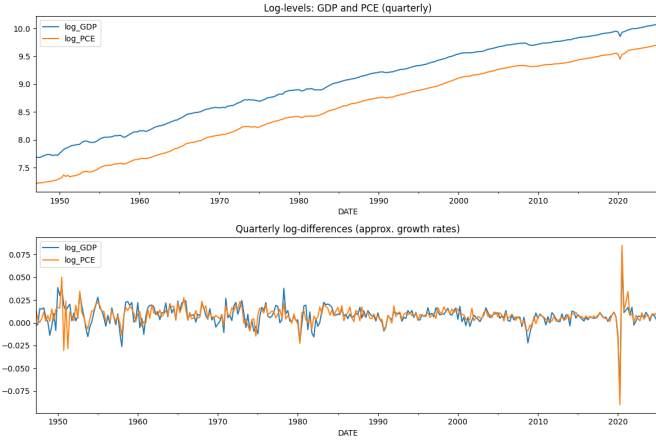


Figure 5: Log-levels of GDP and PCE (top) and quarterly log-differences (bottom).

Interpretation: Both GDP and PCE trend upward in levels, but their first differences fluctuate around zero, suggesting growth rates are stationary. This motivates unit root testing.

Unit Root and Cointegration Tests

Table 1: ADF Unit Root Tests

Series	Stat	p-val	Lag	Obs.	CV(5%)	Result
$\log(GDP)$	-1.51	0.83	1	312	-3.42	$I(1)$
$\log(PCE)$	-1.04	0.94	2	311	-3.42	$I(1)$
$\Delta \log(GDP)$	-15.42	0.00	0	312	-2.87	$I(0)$
$\Delta \log(PCE)$	-8.84	0.00	3	309	-2.87	$I(0)$

Interpretation: Both GDP and PCE are non-stationary in levels ($I(1)$) but stationary in first differences ($I(0)$). This pattern is typical for macroeconomic aggregates and allows the possibility of cointegration [6].

Cointegration and Long-run Relation

We estimate the cointegrating relation by OLS:

$$\log(PCE_t) = -0.995 + 1.060 \log(GDP_t) + \hat{u}_t.$$

Table 2: OLS Regression (Cointegration Step)

Variable	Coeff.	Std.Err.	t-stat	p-val
Const	-0.9946	0.016	-62.91	0.000
$\log(GDP)$	1.0602	0.002	606.18	0.000
R^2	0.999			
DW	0.142			
Obs.	314			

Interpretation: The slope $\hat{\beta} \approx 1.06$ suggests that GDP and consumption move nearly proportionally in the long run. However, residual diagnostics (very low DW statistic) suggest strong autocorrelation, so formal cointegration tests are required [5].

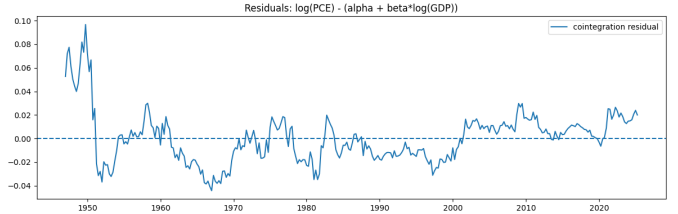


Figure 6: Residuals $\hat{u}_t = \log(PCE_t) - (\hat{\alpha} + \hat{\beta} \log(GDP_t))$ from the cointegration regression.

Interpretation: The residuals fluctuate around zero but show persistent patterns, again pointing to the need for an Engle-Granger test [5].

Engle-Granger Test

Table 3: Engle-Granger Residual Test

Test Stat	p-val	CV(5%)	Result
-4.61	0.0008	-3.36	Stationary

Interpretation: The null of a unit root in residuals is rejected, confirming that GDP and PCE share a long-run equilibrium relationship [5].

Error Correction Model (ECM)

Estimating the short-run correction:

$$\Delta \log(PCE_t) = \rho_0 + \rho_1 \Delta \log(GDP_t) + \gamma \hat{u}_{t-1} + \varepsilon_t.$$

Table 4: ECM Results

Regressor	Coeff.	Std.Err.	t-stat	p-val
Const	0.0024	0.0005	4.96	0.000
$\Delta \log(GDP)_t$	0.7286	0.036	20.05	0.000
$\hat{u}_{t-1}$	-0.0628	0.019	-3.36	0.001
R^2	0.566			
DW	2.41			
Obs.	313			

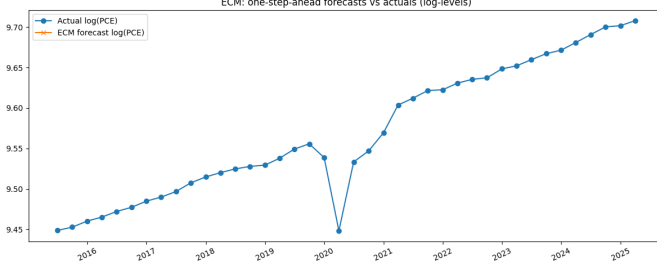


Figure 7: ECM one-step-ahead forecasts vs actuals (log-levels). Forecasts are produced from the estimated ECM.

Interpretation: Consumption growth responds strongly to GDP growth ($\rho_1 \approx 0.73$). The error correction term of -0.063 indicates that about 6% of disequilibrium is corrected each quarter, implying slow but significant adjustment toward long-run equilibrium [5].

Johansen VECM Results

We also run Johansen tests [8] and estimate a VECM system to confirm the rank and obtain the cointegrating vector.

Table 5: Johansen Cointegration Test (Trace)

Null	Trace Stat	CV(5%)	Result
$r = 0$	27.24	15.49	Reject
$r \leq 1$	3.78	3.84	Borderline

Table 6: VECM Estimates

	Coeff.	Std.Err.	p-val
Cointegration vector (β):			
$\log(GDP)$	1.000	—	—
$\log(PCE)$	-0.936	0.006	0.000
Adjustment coeff. (α):			
GDP equation	-0.090	0.031	0.003
PCE equation	0.007	0.030	0.822

Interpretation: The estimated vector suggests GDP and PCE move nearly proportionally in the long run ($\beta \approx$

$(1, -0.936)$). The adjustment coefficients show GDP responds significantly to disequilibria ($\alpha = -0.09$), while PCE is weakly exogenous. This means consumption drives the equilibrium path, and GDP adapts [8].

Diagnosis

Residual diagnostics from ECM (autocorrelation, heteroskedasticity, non-normality) indicate misspecification risk in standard inference.

Table 7: ECM Residual Diagnostics

Test	Stat	p-val
Ljung–Box (12 lags)	47.43	0.0000
Breusch–Pagan	15.60	0.0004
Jarque–Bera	1326.43	0.0000

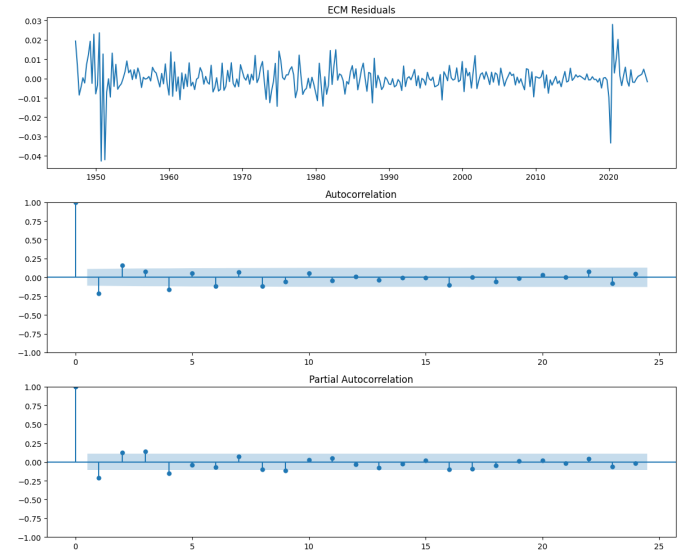


Figure 8: ECM residual diagnostics (ACF/PACF/histogram/QQ or stacked diagnostics as saved).

Interpretation: The ECM residuals show significant autocorrelation, heteroskedasticity, and non-normality. This implies that OLS-based inference may be unreliable, and robust or bootstrap methods are needed [9].

Damage

Without modeling cointegration, OLS regressions would be spurious [6]. Despite cointegration, residual problems remain and may invalidate conventional standard errors and hypothesis tests.

Directions

To improve model reliability:

- Add lags or include additional exogenous variables (e.g., interest rates, unemployment).
- Test for structural breaks/regime changes and estimate split-sample or regime-dependent models.
- Use HAC/clustered standard errors or bootstrap to obtain robust inference.
- Consider ARIMA/V ARIMA with structural breaks or regime-switching VECMs if warranted.

Deployment

ECM/VECM frameworks are useful for macro forecasting, policy analysis, and stress testing. A validated model can produce short-run forecasts of consumption growth and inform long-run equilibrium assessments for fiscal and monetary planning.

Regime Change Challenge: The Case of Delta Airlines (DAL)

Definition

In time series analysis, a **regime change** refers to a structural shift in the data-generating process, where the underlying relationships, such as mean, variance, or dynamics, change over time. These changes often arise from external shocks or policy interventions, and in financial markets they may correspond to transitions between normal, crisis, and recovery periods [10].

Formally, if $\{r_t\}$ denotes asset returns, a regime change implies that the parameters governing r_t differ across segments of the sample:

$$r_t \sim \mathcal{D}(\theta^{(k)}) \quad \text{for } t \in \mathcal{T}_k,$$

where $\mathcal{T}_k$ denotes the k -th regime, and $\theta^{(k)}$ is the parameter vector specific to that regime.

Two common approaches to detecting regime changes are:

- **Structural break tests and segmentation algorithms** (e.g., ruptures), which identify discrete breakpoints in the series where parameters shift [11].
- **Markov Switching Models (MSM)**, where regime changes are governed by an unobserved Markov chain. For the two-regime case:

$$r_t = \mu_{s_t} + \phi_{s_t} r_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_{s_t}^2),$$

with $s_t \in \{1, 2\}$ and transition probabilities

$$P(s_t = j \mid s_{t-1} = i) = p_{ij}, \quad \sum_j p_{ij} = 1.$$

This combination of break detection and regime-switching modelling allows both identification of abrupt structural changes and probabilistic inference about gradual shifts between states.

Description

The model lets returns switch between a low-volatility “normal” regime and a high-volatility “crisis” regime. The timing of switches is probabilistic, making it useful for sudden shocks such as COVID-19 or geopolitical events like the Russia–Ukraine war.

Data Selection

Each team member proposed a dataset for the regime change challenge. The final choice was between Netflix, Delta Airlines, and ExxonMobil.

- **Delta Airlines (DAL)** – COVID breaks: Clear (Feb–Jun 2020) – Industry impact: Direct demand collapse – Clarity: Focused (COVID-driven) – Pedagogical value: High; textbook regime change
- **Netflix (NFLX)** – COVID breaks: Weak, muted signal – Industry impact: Benefited from lockdowns – Clarity: Mixed (growth vs competition) – Pedagogical value: Moderate; less crisis-aligned
- **ExxonMobil (XOM)** – COVID breaks: Present, but mixed with 2023 events – Industry impact: Oil demand shock + later geopolitics – Clarity: Multiple regimes (COVID + Ukraine war) – Pedagogical value: Moderate; confounded by later shocks

We selected **Delta Airlines (DAL)** as the dataset for our regime change analysis. DAL provides the clearest case study of a COVID-induced structural shift, with breaks clustered tightly around February–June 2020. By contrast, NFLX shows weaker signals because streaming benefited during lockdowns, while XOM introduces additional breaks in 2023 linked to geopolitical shocks. DAL therefore offers the most interpretable and pedagogically valuable example of regime change.

Demonstration

We analysed daily Delta Airlines prices (2018–2025). Log returns

$$r_t = 100 \times \ln\left(\frac{P_t}{P_{t-1}}\right)$$

are stationary (ADF $p \approx 0$, KPSS $p = 0.10$), while price levels are not. Using **ruptures** binary segmentation with three breaks, we find structural changes on **19 Feb 2020**, **14 May 2020**, and **5 Jun 2020**. A two-regime MSM fitted to returns shows high crisis probabilities from Feb–Jun 2020, consistent with these breaks.

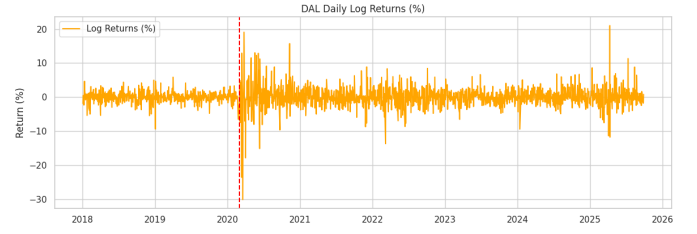
Diagrams

Figure 1: DAL adjusted close with break dates



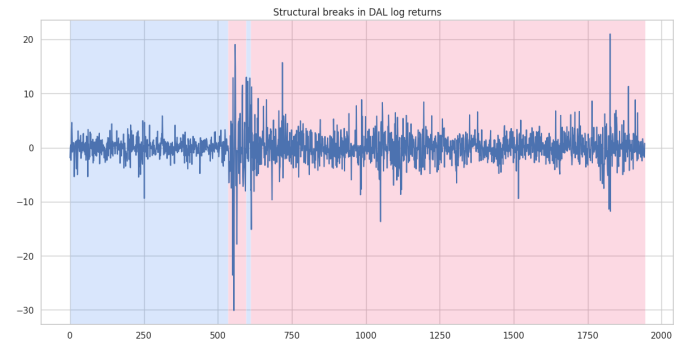
This chart shows that Delta’s stock peaked before the pandemic and then experienced a sharp decline starting in February 2020, with structural breaks marking the onset and stabilization of the crisis.

Figure 2: Structural breaks in DAL daily log returns



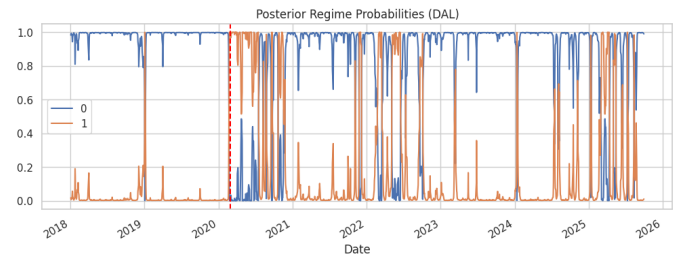
This figure highlights the extreme volatility in daily returns during early 2020, followed by a gradual return to normal variability after June 2020. The vertical lines indicate detected breakpoints, corresponding to major structural changes in volatility and mean levels.

Figure 3: Posterior regime probabilities (MSM)



The Markov–Switching model identifies a high probability of being in the crisis regime from February to June 2020. This confirms that the model successfully captures the COVID-related structural break detected in the returns.

Figure 4: Posterior smoothed probabilities for two regimes



This plot shows the estimated probability of being in each regime over time. The high-volatility crisis regime (orange) dominates from February to mid-2020, coinciding with the COVID-19 shock, before gradually giving way to the low-volatility regime (blue). Later fluctuations occur but do not persist as strongly as the initial crisis episode.

Diagnosis

The structural break analysis and the Markov–Switching model produce results that are consistent with each other.

Both methods highlight a transition into a high-volatility state in late February 2020, coinciding with the onset of the COVID-19 shock to the airline industry, and an exit into a more stable regime by mid-2020. The alignment of methods provides strong evidence that the observed breaks are not statistical noise but reflect genuine structural changes. Moreover, the regime probability plot confirms that the crisis regime persisted for several months before reverting, which matches the economic reality of grounded flights and gradual reopening.

Damage

Despite these strengths, several limitations must be acknowledged. First, the price series is non-stationary, meaning that modelling levels directly could lead to misleading conclusions. Second, structural break detection is sensitive to the number of breaks specified; different penalty parameters could either overfit or miss important events. Third, the use of only two regimes in the MSM may oversimplify the dynamics: in practice, the crisis unfolded in multiple phases (initial collapse, trough, and recovery). Finally, both approaches assume relatively clean data, but extreme outliers in returns could bias break detection and parameter estimates.

Directions

Future work can address these limitations in several ways. Allowing for three or more regimes would capture distinct phases of the pandemic, such as stabilization and recovery, rather than grouping them into a single high-volatility state. Alternatively, restricting the sample to 2018–2021 would sharpen focus on the pandemic without interference from later shocks. Incorporating GARCH-type errors within the Markov–Switching framework could also improve modelling of volatility clustering, which is a prominent feature of financial returns. Robustness checks, such as trimming extreme outliers or using alternative break detection algorithms, would further strengthen confidence in the results.

Deployment

In practice, regime-switching models and break detection tools can serve as early-warning indicators. Investors could monitor the smoothed probability of crisis regimes and reduce exposure when this probability exceeds a threshold, thereby mitigating downside risk. Airlines could use similar models to stress-test revenues under different volatility regimes and adjust financing or capacity planning accordingly. Policymakers may also benefit: identifying regime shifts in airline data provides a proxy for real-time monitoring of the broader economy, since air travel demand is closely linked to business activity and consumer confidence.

References

- [1] Bishop, C. M. *Pattern Recognition and Machine Learning*. Springer, 2006.
- [2] Tsay, R. S. *Analysis of Financial Time Series*. 3rd ed., Wiley, 2010.
- [3] Jolliffe, I. T. *Principal Component Analysis*. 2nd ed., Springer, 2002.
- [4] Chan, L. K. C., Karceski, J., and Lakonishok, J. “On Portfolio Optimization: Forecasting Covariances and Choosing the Risk Model.” *Review of Financial Studies*, 2007.
- [5] Engle, R. F., & Granger, C. W. J. (1987). Cointegration and error correction: Representation, estimation, and testing. *Econometrica*, 55(2), 251–276.
- [6] Granger, C. W. J., & Newbold, P. (1974). Spurious regressions in econometrics. *Journal of Econometrics*, 2(2), 111–120.
- [7] Hamilton, J. D. (1994). *Time Series Analysis*. Princeton University Press.
- [8] Johansen, S. (1991). Estimation and hypothesis testing of cointegration vectors in Gaussian vector autoregressive models. *Econometrica*, 59(6), 1551–1580.
- [9] Stock, J. H., & Watson, M. W. (2019). *Introduction to Econometrics* (4th ed.). Pearson.
- [10] Hamilton, James D. “A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle.” *Econometrica*, vol. 57, no. 2, 1989, pp. 357–384.
- [11] Truong, Charles, Laurent Oudre, and Nicolas Vayatis. “Selective Review of Offline Change Point Detection Methods.” *Signal Processing*, vol. 167, 2020.